Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 1 / 方法 1 [NT08-M01]

Divisor functions are products of local exponent choices. Learn the formulas together with their inverse patterns, parity exceptions, coprimality restrictions and exact tail bounds; these are what make the formulas useful in contest questions.

约数函数是局部指数选择的乘积。公式须与反向指数模式、奇偶例外、互质限制及精确尾项界一起掌握，才能在竞赛题中真正使用。

Count and sum divisors by independent prime exponents

以独立质指数统计约数及约数和

Procedure

1. Factor $N = \prod p^{e_p}$ and list each exponent choice $0, 1, \ldots, e_p$.

2. Use $\tau(N) = \prod(e_p + 1)$ and $\sigma(N) = \prod(1 + p + \cdots + p^{e_p})$.

3. If only certain exponents are allowed, replace each local factor by that permitted count or sum.

4. Only after the exact divisor count is known should it be inserted into a probability or a later equation.

通用步骤

1. 分解 $N = \prod p^{e_p}$，每个质数指数可选 $0, 1, \ldots, e_p$。

2. 使用 $\tau(N) = \prod(e_p + 1)$ 与 $\sigma(N) = \prod(1 + p + \cdots + p^{e_p})$。

3. 若只允许某些指数，将每个局部因子替换为允许指数的个数或幂和。

4. 明确精确约数数目之后，才代入概率或后续方程。

> Example. For $N = 2^4 3^2$, count its square divisors and find their sum.

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Method 1 / 方法 1 [NT08-M01] (continued / 续)

> 例题：对 $N = 2^4 3^2$，求平方约数的个数及其总和。

Worked solution

1. Square exponents are $0, 2, 4$ for two and $0, 2$ for three, so there are $3 \cdot 2 = 6$ square divisors.

2. The sum factors as $(1 + 2^2 + 2^4)(1 + 3^2)$.

3. The value is $21 \cdot 10 = 210$.

例题解析

1. 二的平方指数为 $0, 2, 4$，三的为 $0, 2$，所以共有 $3 \cdot 2 = 6$ 个平方约数。

2. 其和分解为 $(1 + 2^2 + 2^4)(1 + 3^2)$。

3. 结果为 $21 \cdot 10 = 210$。

Restrictions belong on exponents; listing divisors is usually unnecessary.

限制应施加于指数，通常无需逐个列约数。

Practice / 配套练习: AIME-2005-I-Q02 (WB p. 600), AIME-2010-I-Q01 (WB p. 723), AIME-2011-I-Q07 (WB p. 498), AIME-2019-II-Q03 (WB p. 487), AIME-2023-I-Q04 (WB p. 473), AIME-2023-II-Q10 (WB p. 716), AIME-2026-I-Q06 (WB p. 124), AIME-2026-I-Q08 (WB p. 507)

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约数函数与欧拉函数

Method 2 / 方法 2 [NT08-M02]

Invert a divisor count into prime-exponent patterns

将约数个数反解为质指数模式

Procedure

1. Write the required count as a product of factors $e_i + 1 > 1$.

2. List every multiplicative partition, then impose prime restrictions and numerical bounds.

3. To minimize an integer with fixed exponents, assign larger exponents to smaller primes; swapping proves this rule.

4. For constrained divisors of a large power, check that every recovered exponent lies in the original range.

通用步骤

1. 将指定约数个数写成若干 $e_i + 1 > 1$ 的乘积。

2. 列出全部乘法拆分，再施加质数限制与数值界。

3. 固定指数求最小整数时，将较大指数配较小质数；交换可证明此规则。

4. 对大幂的受限约数，核对每个还原指数均在原范围内。

> Example. Find the least positive integer with exactly eight positive divisors.

> 例题：求恰有八个正约数的最小正整数。

Worked solution

1. The exponent patterns are 7, $(3, 1)$ and $(1, 1, 1)$, from 8, $4 \cdot 2$ and $2 \cdot 2 \cdot 2$.

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约数函数与欧拉函数

Method 2 / 方法 2 [NT08-M02] (continued / 续)

2. Their least values are $2^7 = 128$, $2^3 \cdot 3 = 24$ and $2 \cdot 3 \cdot 5 = 30$.

3. The minimum is 24, with divisor count $(3+1)(1+1) = 8$.

例题解析

1. 由 $8, 4 \cdot 2, 2 \cdot 2 \cdot 2$ 得指数模式 $7, (3,1), (1,1,1)$。

2. 各模式最小值为 $2^7 = 128, 2^3 \cdot 3 = 24, 2 \cdot 3 \cdot 5 = 30$。

3. 最小值为 24，其约数数为 $(3+1)(1+1) = 8$。

Include repeated exponents and the case of a single prime power.

须包含重复指数与只有一个质数幂的情形。

For consecutive integers with a small prescribed sum of divisor counts, classify the possible count pairs first. A count of three forces a prime square, five forces a prime fourth power, and four forces p^3 or pq with distinct primes p, q. This turns a long integer search into a bounded search around prime powers.

相邻整数的约数数目之和为小定值时，先分类可能的计数对。约数数为三迫使质数平方，为五迫使质数四次幂，为四迫使 p^3 或不同质数 p, q 的乘积 pq。这能把很长的整数搜索化为质数幂附近的有界搜索。

Practice / 配套练习: AIME-2000-II-Q04 (WB p. 599), AIME-2004-II-Q08 (WB p. 605), AIME-2005-I-Q03 (WB p. 601), AIME-2006-II-Q13 (WB p. 610), AIME-2019-I-Q09 (WB p. 606), AIME-2019-II-Q09 (WB p. 607)

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约数函数与欧拉函数

Method 3 / 方法 3 [NT08-M03]

Characterize divisor counts of kth powers

刻画 k 次幂的约数个数

Procedure

1. If $a = \prod p_i^{e_i}$, then $\tau(a^k) = \prod(ke_i + 1)$.

2. Every factor is one modulo k, giving a necessary congruence on the divisor count.

3. To prove sufficiency, realize a count $N = ke + 1$ using a single prime power $a = p^e$, including $e = 0$.

通用步骤

1. 若 $a = \prod p_i^{e_i}$，则 $\tau(a^k) = \prod(ke_i + 1)$。

2. 每个因子都模 k 余一，从而得到约数个数的必要同余。

3. 证明充分性时，用单一质数幂 $a = p^e$ 实现个数 $N = ke + 1$，并包括 $e = 0$。

> Example. Which positive integers can be the number of divisors of an integer fourth power?

> 例题：哪些正整数可以是某整数四次幂的正约数个数?

Worked solution

1. Every such count is $\prod(4e_i + 1) \equiv 1 \pmod 4$.

2. Conversely, for $N = 4e + 1$, the fourth power of 2^e has exactly $4e + 1 = N$ divisors.

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约数函数与欧拉函数

Method 3 / 方法 3 [NT08-M03] (continued / 续)

3. Thus exactly the positive integers congruent to one modulo four occur, including one from the integer one.

例题解析

1. 每个这样的个数均为 $\prod(4e_i+1)\equiv 1 \pmod 4$。

2. 反之，对 $N=4e+1$，2^e 的四次幂恰有 $4e+1=N$ 个约数。

3. 所以恰为模四余一的正整数，包括整数一产生的个数一。

> Necessity from a product congruence needs a separate realization for sufficiency.

> 乘积同余给出的必要性，须另给实现才能证明充分性。

Practice / 配套练习: AIME-2016-II-Q11 (WB p. 611)

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约数函数与欧拉函数

Method 4 / 方法 4 [NT08-M04]

Count coprime classes with the totient and a symmetry pairing

用欧拉函数与对称配对统计互质类

Procedure

1. For $N = \prod p^e$, inclusion-exclusion gives $\varphi(N) = N \prod_{p \mid N} (1 - 1/p)$.

2. When an interval contains full residue periods, count those periods and then the boundary remainder.

3. Before halving a count by $a \leftrightarrow N - a$, prove the map has no fixed admissible element.

4. If some primes may divide the numerator, partition by those primes and apply the largest permitted coprime denominator bound.

通用步骤

1. 对 $N = \prod p^e$，容斥给出 $\varphi(N) = N \prod_{p \mid N} (1 - 1/p)$。

2. 区间含完整剩余周期时，先数完整周期，再处理边界余段。

3. 用 $a \leftrightarrow N - a$ 将个数减半前，证明没有合格不动点。

4. 若某些质数允许整除分子，按这些质数分类，并施加最大允许互质分母界。

> Example. Choose a residue uniformly from $0, \ldots, 29$. Given that it is odd, what is the probability it has a multiplicative inverse modulo 30?

> 例题：从 $0, \ldots, 29$ 等可能选一个剩余。已知它是奇数，其模 30 可逆的概率是多少？

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约数函数与欧拉函数

Method 4 / 方法 4 [NT08-M04] (continued / 续)

Worked solution

1. Invertibility means coprimality with $30 = 2 \cdot 3 \cdot 5$.

2. There are $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8$ invertible residues, all odd.

3. The conditioning leaves fifteen odd residues, so the probability is $8/15$.

例题解析

1. 可逆等价于与 $30 = 2 \cdot 3 \cdot 5$ 互质。

2. 可逆剩余有 $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8$ 个，且全为奇数。

3. 条件下共有十五个奇数剩余，故概率为 $8/15$。

Totient counts all coprime residues, not automatically only half of an interval.

欧拉函数统计全部互质剩余，并非自动给出某区间的一半。

Practice / 配套练习: AIME-2004-I-Q08 (WB p. 604), AIME-2014-I-Q03 (WB p. 603), AIME-2022-I-Q13 (WB p. 598)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 5 / 方法 5 [NT08-M05]

Use divisor pairing to track parity through square blocks

用约数配对按平方区块追踪奇偶

Procedure

1. Pair every divisor d with n/d; an unpaired divisor exists exactly when n is square.

2. Therefore $\tau(n)$ is odd exactly at squares, and a cumulative divisor-count sum changes parity only there.

3. Count entire intervals between consecutive squares and then the final partial interval.

通用步骤

1. 把每个约数 d 与 n/d 配对；仅平方数有未配对约数。

2. 因此 $\tau(n)$ 恰在平方数时为奇数，累积约数计数和也仅在此改变奇偶。

3. 统计连续平方数之间的完整区间，再处理最后不完整区间。

> Example. Among $1 \le n \le 20$, for how many n is $\sum_{j=1}^{n} \tau(j)$ odd?

> 例题：在 $1 \le n \le 20$ 中，有多少个 n 使 $\sum_{j=1}^{n} \tau(j)$ 为奇数?

Worked solution

1. The parity is the parity of the number of squares at most n, namely $\lfloor \sqrt{n} \rfloor$.

2. Odd square-root floors occur on $1 \le n \le 3$ and $9 \le n \le 15$.

3. These contain $3 + 7 = 10$ integers.

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约数函数与欧拉函数

Method 5 / 方法 5 [NT08-M05] (continued / 续)

例题解析

1. 其奇偶等于不超过 n 的平方数数目，即 $\lfloor\sqrt{n}\rfloor$ 的奇偶。

2. 平方根下取整为奇数的区间是 $1 \le n \le 3$ 与 $9 \le n \le 15$。

3. 共含 $3+7=10$ 个整数。

A square switches the cumulative parity at the square itself.

累积奇偶从平方数本身起切换。

Practice / 配套练习: AIME-2005-I-Q12 (WB p. 609)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 6 / 方法 6 [NT08-M06]

Intersect divisor sets with gcds and use multiplicativity carefully

以最大公因数求约数集交，并谨慎使用积性

Procedure

1. Divisors common to several integers are exactly divisors of their gcd.

2. Count unions by inclusion-exclusion, including the divisor one in every intersection.

3. For σ or τ of a product, multiply values only when the factors are coprime, or merge their prime exponents first.

通用步骤

1. 若干整数的公共约数恰为其最大公因数的约数。

2. 用容斥统计并集，并在每个交集中包括约数一。

3. 乘积的 σ 或 τ 只有在因子互质时才可相乘，否则须先合并质指数。

> Example. How many positive integers divide 18 or 24?

> 例题：有多少正整数整除 18 或 24?

Worked solution

1. $\tau(18) = (1+1)(2+1) = 6$ and $\tau(24) = (3+1)(1+1) = 8$.

2. The intersection is the divisor set of $\gcd(18, 24) = 6$, containing $\tau(6) = 4$ values.

3. The union count is $6 + 8 - 4 = 10$.

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约数函数与欧拉函数

Method 6 / 方法 6 [NT08-M06] (continued / 续)

例题解析

1. $\tau(18)=(1+1)(2+1)=6, \tau(24)=(3+1)(1+1)=8$。

2. 交集为 $\gcd(18,24)=6$ 的约数集，有 $\tau(6)=4$ 个值。

3. 并集个数为 $6+8-4=10$。

The functions are multiplicative on coprime inputs, not on every product.

这些函数只对互质输入具有积性，并非对任意乘积都可相乘。

Practice / 配套练习: AIME-2005-II-Q04 (WB p. 602), AIME-2021-I-Q14 (WB p. 612)

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约数函数与欧拉函数

Method 7 / 方法 7 [NT08-M07]

Regroup a positive series into divisor coefficients and bound its tail

把正项级数重组为约数系数，并估计尾项

Procedure

1. Expand each geometric denominator into a nonnegative series. Nonnegativity justifies regrouping by exponent.

2. An exponent n receives one contribution for each factorization $n = jk$, giving coefficient $\tau(n)$.

3. After scaling by a base power, separate the integral prefix from the tail and prove the tail is below the next integer before reading digits.

4. Use an elementary bound such as $\tau(n) \le n$ for distant terms, while calculating the first few tail coefficients exactly.

通用步骤

1. 把每个等比分母展开为非负级数；非负性允许按指数重组。

2. 指数 n 对每种 $n = jk$ 的分解得到一次贡献，因此系数为 $\tau(n)$。

3. 乘以底数幂后，将整数前缀与尾项分开；读取数位之前先证明尾项未达到下一整数。

4. 对远处项使用 $\tau(n) \le n$ 等简单界，对尾项开头若干系数则精确计算。

> Example. For $T = \sum_{k \ge 1} 1/(100^k - 1)$, identify the coefficient of 100^{-6} and prove $T < 1$.

> 例题：对 $T = \sum_{k \ge 1} 1/(100^k - 1)$，求 100^{-6} 的系数，并证明 $T < 1$。

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约数函数与欧拉函数

Method 7 / 方法 7 [NT08-M07] (continued / 续)

Worked solution

1. Expand $1/(100^k-1)=\sum_{j\ge1}100^{-kj}$. Exponent six arises from $k=1,2,3,6$, so its coefficient is four.

2. Thus $T=\sum_{n\ge1}\tau(n)100^{-n}\le\sum_{n\ge1}n100^{-n}$.

3. The last sum equals $(1/100)/(1-1/100)^2=100/9801<1$.

例题解析

1. 展开 $1/(100^k-1)=\sum_{j\ge1}100^{-kj}$。指数六来自 $k=1,2,3,6$，故系数为四。

2. 所以 $T=\sum_{n\ge1}\tau(n)100^{-n}\le\sum_{n\ge1}n100^{-n}$。

3. 最后一和为 $(1/100)/(1-1/100)^2=100/9801<1$。

Coefficients exceeding the base may carry; a series coefficient is not automatically a digit.

系数超过底数时会进位，级数系数并不自动等于数位。

Practice / 配套练习: AIME-2026-II-Q09 (WB p. 608)